

Benchmark Problem (Extrapolating the Velocities at Ghost Points)

```
%-----Domain Parameter-----%
% Grid Points
global Nx; Nx = 106;
global Ny; Ny = 56;

%--Important Parameter
INFINITE = 10^12;
eta = 10^-6;
error_eta = 10^-9;
CFL = 0.5;

% To track the error for each iterations
iteration_error = [];

% Dimension of Domain
Lx = 100;
Ly = 50;

% Domain
x = linspace(0,Lx,Nx);
y = linspace(0,Ly,Ny);
[X,Y] = meshgrid(x,y);

% Defining velocity
global speed discomfort cost x_velocity y_velocity;

% Space Interval
global h;
h = Lx/(Nx-6);

% Total Time
t_total = 250;

% --- Obstacle modeling ----%
global obstacle;
obstacle = false(Ny,Nx);
obstacle(14:33,44:63) = true;

%-----Initial Condition-----%
% Zero Density initial condition
rho_zeros = zeros(Ny,Nx);

rho = rho_zeros;

%--Function to compute Velocity using Density
function speed = makeSpeed(rho)
    global Nx Ny;
```

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    speed = zeros(Ny,Nx);
    for i = 1:1:Nx
        for j = 1:1:Ny
            speed(j,i) = 2*(1 - rho(j,i)/10);
        end
    end
end

%--Function to compute Discomfort using Density
function discomfort = makeDiscomfort(rho)
    global Nx Ny;
    discomfort = zeros(Ny,Nx);
    for i = 1:1:Nx
        for j = 1:1:Ny
            discomfort(j,i) = 0.002*rho(j,i)*rho(j,i);
        end
    end
end

%--Function to compute Cost function using Speed and Discomfort
function cost = makeCost(speed,discomfort)
    global Nx Ny;
    cost = zeros(Ny,Nx);
    for i = 1:1:Nx
        for j = 1:1:Ny
            cost(j,i) = 1/speed(j,i) + discomfort(j,i);
        end
    end
end

speed = makeSpeed(rho);
discomfort = makeDiscomfort(rho);
cost = makeCost(speed,discomfort);

% Plotting Speed
figure(1)
contourf(X,Y,speed,'LineStyle', 'none');

```

Warning: Contour not rendered for constant ZData

```

title('Speed (t=0)');
clim([0,2]);
axis equal tight;
colorbar;
xlabel('x'); ylabel('y');
% Plotting Discomfort
figure(2)
contourf(X,Y,discomfort,'LineStyle', 'none');

```

Warning: Contour not rendered for constant ZData

```

title('Discomfort (t=0)');
axis equal tight;
colorbar;
xlabel('x'); ylabel('y');
% Plotting Cost Function
figure(3);
contourf(X,Y,cost,'LineStyle', 'none');
title('Cost Function (t=0)');
axis equal tight;
colorbar;
xlabel('x'); ylabel('y');
% Plotting the Initial Condition
figure(4);
contourf(X,Y,rho,'LineStyle', 'none');
title(' Density (t = 0)');
clim([0,1]);
axis equal tight;
colorbar;
xlabel('x'); ylabel('y');
% Function that will calculate the flux depending on the u,v and Density.
% X_velocity and y_velocity will be found using the potential.
function flux = makeFlux(rho,x_velocity,y_velocity)
    global Nx Ny;

    % Initializing Flux
    flux_x = zeros(Ny,Nx);
    flux_y = zeros(Ny,Nx);
    flux = zeros(2, Ny, Nx);

    for i = 1:1:Nx
        for j = 1:1:Ny

            flux_x(j,i) = rho(j,i) * x_velocity(j,i);
            flux_y(j,i) = rho(j,i) * y_velocity(j,i);

            flux(1,j,i) = flux_x(j,i);
            flux(2,j,i) = flux_y(j,i);
        end
    end
end

%--Pre allocating the memory
rho_n = zeros(Ny,Nx);
rho_1 = zeros(Ny,Nx);
rho_2 = zeros(Ny,Nx);
rho_next = zeros(Ny,Nx);

% Stopping condition
stop_low = -1;
stop_high = 11;

```

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% This will be used for plotting
x_plot = linspace(0, Lx, Nx-6);
y_plot = linspace(0, Ly, Ny-6);
[~, ix] = min(abs(x_plot - 36));    % index closest to x = 36
[~, iy] = min(abs(y_plot - 33));    % index closest to y = 33

% This will be used to store the results of the rho values at slices
rho_at_120s = zeros(Ny-6,Nx-6);
rho_at_36 = zeros(Ny-6,1);
rho_at_33 = zeros(1,Nx-6);

phi_at_120 = zeros(Ny-6,Nx-6);
Vx = zeros(Ny-6,Nx-6);
Vy = zeros(Ny-6,Nx-6);

capture_time = 0;

%-----MAIN SOLVER-----%
t = 0;
step = 0; % For plotting at regular interval

% Video Capturing
rho_max = 10;    % z-axis cap for the surface (matches clim([0 10]) used in the
contour)
vel_max = 2;

% ---- SEPARATE FIGURES, ONE PER PLOT ----
figContour = figure(6); set(figContour, 'Position', [50, 500, 700, 500]);
axContour = axes(figContour);

figSurface = figure(7); set(figSurface, 'Position', [800, 500, 700, 500]);
axSurface = axes(figSurface);

figVelocity = figure(8); set(figVelocity, 'Position', [50, 0, 700, 500]);
axVelocity = axes(figVelocity);

vContour = VideoWriter('Density_Contour.mp4', 'MPEG-4'); vContour.FrameRate =
30; open(vContour);
vSurface = VideoWriter('Density_Surface.mp4', 'MPEG-4'); vSurface.FrameRate =
30; open(vSurface);
vVelocity = VideoWriter('Velocity_Surface.mp4', 'MPEG-4'); vVelocity.FrameRate =
30; open(vVelocity);

while t < t_total

    % Computing Cost For the density rho
    speed = makeSpeed(rho);
    discomfort = makeDiscomfort(rho);

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cost = makeCost(speed,discomfort);

%-- Calculate Initial Phi guess using Lower order Fast Sweep
phi_first_order = First_order_Godunov_Fast_Sweep_Algorithm(cost,h,INFINITE);
%-- Calculate The final phi using higher order scheme and above initial guess
phi =
weno_fast sweep(cost,h,eta,error_eta,INFINITE,phi_first_order,iteration_error);

% Extrapolation of the phi values
% Bottom Edge (3 Ghost Points)
phi(3,:) = phi(4,:);
phi(2,:) = phi(3,:);
phi(1,:) = phi(2,:);
% Top Edge (3 Ghost Points)
phi(Ny-2,:) = phi(Ny-3,:);
phi(Ny-1,:) = phi(Ny-2,:);
phi(Ny,:) = phi(Ny-1,:);
% Right Edge (3 Ghost Points)
for j = 1:Ny
    if (j >= 9 && j <= 23) || (j >= 34 && j <= 48) % exit rows: linear
extrapolation
        slope = phi(j,Nx-3) - phi(j,Nx-4);
        phi(j,Nx-2) = phi(j,Nx-3) + slope;
        slope = phi(j,Nx-4) - phi(j,Nx-5);
        phi(j,Nx-1) = phi(j,Nx-2) + slope;
        slope = phi(j,Nx-5) - phi(j,Nx-6);
        phi(j,Nx) = phi(j,Nx-1) + slope;
    else % wall rows: keep flat
copy
        phi(j,Nx-2) = phi(j,Nx-3);
        phi(j,Nx-1) = phi(j,Nx-2);
        phi(j,Nx) = phi(j,Nx-1);
    end
end
% Left Edge (3 Ghost Points)
phi(:,3) = phi(:,4);
phi(:,2) = phi(:,3);
phi(:,1) = phi(:,2);

%--Calculate the Components of Velocity
[phi_x, phi_y] = gradient(phi, h, h); % partial derivatives of phi in x and y
mag_phi = sqrt(phi_x.^2 + phi_y.^2);
mag_phi = max(mag_phi, eta); % To be safe with denominator

for i = 1:1:Nx
    for j = 1:1:Ny
        x_velocity(j,i) = speed(j,i) * (- phi_x(j,i))/mag_phi(j,i);
        y_velocity(j,i) = speed(j,i) * (- phi_y(j,i))/mag_phi(j,i);
    end
end

```

```

% Updating the delta_t with proper CFL condition (Stochastic paper)
u_max = max(abs(x_velocity(:)));
v_max = max(abs(y_velocity(:)));
%dt = CFL / ( ( u_max/h ).^(5/3) + ( v_max/h ).^(5/3) + 1e-12 );
dt = CFL * h / ( ( u_max ) + ( v_max ) + 1e-12 );

% -- Making rho_n and RHS of rho_n
rho_n = rho;

semi_discritze_form_rho_n = makeRHS(rho_n,t,speed);

% Check for the negative rho_n values
if any(rho_n(:) < stop_low)
    fprintf('step = %d\n',step);
    error('Negative density detected at rho_n!');
elseif any(rho_n(:) > stop_high)
    fprintf('step = %d\n',step);
    error('Density more than 10 detected at rho_n!');
end

% -- Extrapolation (My Intuition)
% Left Edge      (3 Ghost Points)
rho_n(:,3) = rho_n(:,4);
rho_n(:,2) = rho_n(:,3);
rho_n(:,1) = rho_n(:,2);
% Bottom Edge    (3 Ghost Points)
rho_n(3,:) = rho_n(4,:);
rho_n(2,:) = rho_n(3,:);
rho_n(1,:) = rho_n(2,:);
% Top Edge       (3 Ghost Points)
rho_n(Ny-2,:) = rho_n(Ny-3,:);
rho_n(Ny-1,:) = rho_n(Ny-2,:);
rho_n(Ny,:) = rho_n(Ny-1,:);
% Right Edge     (3 Ghost Points)
rho_n(:,Nx-2) = rho_n(:,Nx-3);
rho_n(:,Nx-1) = rho_n(:,Nx-2);
rho_n(:,Nx) = rho_n(:,Nx-1);

% ----- Making the RHO_1 and then RHS of rho_1 ----- %
for i = 4:1:Nx-3
    for j = 4:1:Ny-3
        if(~obstacle(j,i))
            rho_1(j,i) = rho_n(j,i) + dt * semi_discritze_form_rho_n(j-3,i-3);
        else
            rho_1(j,i) = 0;
        end
    end
end

```

```

end

% -- Extrapolation (My Intuition)
% Left Edge      (3 Ghost Points)
rho_1(:,3) = rho_1(:,4);
rho_1(:,2) = rho_1(:,3);
rho_1(:,1) = rho_1(:,2);
% Bottom Edge    (3 Ghost Points)
rho_1(3,:) = rho_1(4,:);
rho_1(2,:) = rho_1(3,:);
rho_1(1,:) = rho_1(2,:);
% Top Edge       (3 Ghost Points)
rho_1(Ny-2,:) = rho_1(Ny-3,:);
rho_1(Ny-1,:) = rho_1(Ny-2,:);
rho_1(Ny,:) = rho_1(Ny-1,:);
% Right Edge     (3 Ghost Points)
rho_1(:,Nx-2) = rho_1(:,Nx-3);
rho_1(:,Nx-1) = rho_1(:,Nx-2);
rho_1(:,Nx) = rho_1(:,Nx-1);

% -- RHS of discretized form using rho_1
semi_discretize_form_rho_1 = makeRHS(rho_1,t,speed);

% Check for the negative rho_1 values
if any(rho_n(:) < stop_low)
    fprintf('step = %d\n',step);
    error('Negative density detected at rho_1!');
elseif any(rho_n(:) > stop_high)
    fprintf('step = %d\n',step);
    error('Density more than 10 detected at rho_1!');
end

%----- Making the RHO_2 and RHS of RHO_2 -----%
for i = 4:1:Nx-3
    for j = 4:1:Ny-3
        if(~obstacle(j,i))
            rho_2(j,i) = 3/4 * rho_n(j,i) + 1/4 * (rho_1(j,i) + dt *
semi_discretize_form_rho_1(j-3,i-3));
        else
            rho_2(j,i) = 0;
        end
    end
end

end

% -- Extrapolation (My Intuition)

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% Left Edge      (3 Ghost Points)
rho_2(:,3) = rho_2(:,4);
rho_2(:,2) = rho_2(:,3);
rho_2(:,1) = rho_2(:,2);
% Bottom Edge    (3 Ghost Points)
rho_2(3,:) = rho_2(4,:);
rho_2(2,:) = rho_2(3,:);
rho_2(1,:) = rho_2(2,:);
% Top Edge       (3 Ghost Points)
rho_2(Ny-2,:) = rho_2(Ny-3,:);
rho_2(Ny-1,:) = rho_2(Ny-2,:);
rho_2(Ny,:) = rho_2(Ny-1,:);
% Right Edge     (3 Ghost Points)
rho_2(:,Nx-2) = rho_2(:,Nx-3);
rho_2(:,Nx-1) = rho_2(:,Nx-2);
rho_2(:,Nx) = rho_2(:,Nx-1);

% -- RHS of discretized form using rho_2
semi_discretize_form_rho_2 = makeRHS(rho_2,t,speed);

% Check for the negative rho_2 values
if any(rho_n(:) < stop_low)
    fprintf('step = %d\n',step);
    error('Negative density detected at rho_2!');
elseif any(rho_n(:) > stop_high)
    fprintf('step = %d\n',step);
    error('Density more than 10 detected at rho_2!');
end

%-----Making the rho_next-----%

for i = 4:1:Nx-3
    for j = 4:1:Ny-3
        if(~obstacle(j,i))
            rho_next(j,i) = 1/3 * rho_n(j,i) + 2/3 * (rho_2(j,i) + dt *
semi_discretize_form_rho_2(j-3,i-3));
        else
            rho_next(j,i) = 0;
        end
    end
end

% -- Extrapolation (My Intuition)
% Left Edge      (3 Ghost Points)
rho_next(:,3) = rho_next(:,4);
rho_next(:,2) = rho_next(:,3);
rho_next(:,1) = rho_next(:,2);
% Bottom Edge    (3 Ghost Points)
rho_next(3,:) = rho_next(4,:);
rho_next(2,:) = rho_next(3,:);

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rho_next(1,:) = rho_next(2,:);
% Top Edge      (3 Ghost Points)
rho_next(Ny-2,:) = rho_next(Ny-3,:);
rho_next(Ny-1,:) = rho_next(Ny-2,:);
rho_next(Ny,:) = rho_next(Ny-1,:);
% Right Edge    (3 Ghost Points)
rho_next(:,Nx-2) = rho_next(:,Nx-3);
rho_next(:,Nx-1) = rho_next(:,Nx-2);
rho_next(:,Nx) = rho_next(:,Nx-1);

% Substituting back the rho value
rho = rho_next;

%--Check if any of the density values is equal to zero
if any(rho_n(:) < stop_low)
    fprintf('step = %d\n',step);
    error('Negative density detected at rho_next!');
elseif any(rho_n(:) > stop_high)
    fprintf('step = %d\n',step);
    error('Density more than 10 detected at rho_next!');
end

% Slice of density at y = 25
rho_slice = rho_next(25,4:Nx-3);

%--Plotting the rho_next;
if mod(step, 2) == 0

    % To mask the Obstacle
    rho_plot = rho(4:Ny-3, 4:Nx-3);
    rho_plot(14:33,44:63) = NaN;

    speed_plot = speed(4:Ny-3, 4:Nx-3);
    speed_plot(14:33,44:63) = NaN;

    % ----- Contour -----
    contourf(axContour, x_plot, y_plot, rho(4:Ny-3, 4:Nx-3), 150, 'LineStyle',
'none');
    xlim(axContour, [0 Lx]); ylim(axContour, [0 Ly]);
    colorbar(axContour); clim(axContour, [0 10]); colormap(axContour, jet(256));
    title(axContour, sprintf('Density (rho) at t = %.2f', t));
    xlabel(axContour, 'x'); ylabel(axContour, 'y');
    axis(axContour, 'equal', 'tight');

    % ----- Density Surface -----
    surf(axSurface, x_plot, y_plot, rho_plot, 'EdgeColor', 'black',
'FaceColor', 'none');
    view(axSurface, 45+90, 30);
    title(axSurface, sprintf('Density Surface (rho) at t=%.2f', t));

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xlabel(axSurface, 'x'); ylabel(axSurface, 'y');
zlim(axSurface, [-0.5 10.5]);
pbaspect(axSurface, [2 1 2]);

% ----- Velocity Surface -----
surf(axVelocity, x_plot, y_plot, speed_plot, 'EdgeColor', 'black',
'FaceColor', 'none');
view(axVelocity, 45+90, 30);
title(axVelocity, sprintf('Speed Surface at t=%.2f', t));
xlabel(axVelocity, 'x'); ylabel(axVelocity, 'y');
zlim(axVelocity, [-0.5 2.5]);
pbaspect(axVelocity, [2 1 1]);

drawnow;

% Capture the WHOLE FIGURE, not just the axes
writeVideo(vContour, getframe(figContour));
writeVideo(vSurface, getframe(figSurface));
writeVideo(vVelocity, getframe(figVelocity));

% Store the plot of the rho slice at x = 36m at t = 120 sec.
if abs(t - 120) <= 0.02 % tolerance
    % Density Values at 120s
    rho_at_120s = rho(4:Ny-3,4:Nx-3); % For Density contour at 120s
    rho_at_36 = rho_at_120s(:, ix); % For Density slice at 120s
    rho_at_33 = rho_at_120s(iy, :); % For Density slice at 120s
    % Potential Surface at 120s
    phi_at_120 = phi(4:Ny-3, 4:Nx-3);
    % velocity Component at 120s
    Vx_at_120 = x_velocity(4:Ny-3, 4:Nx-3);
    Vy_at_120 = y_velocity(4:Ny-3, 4:Nx-3);

    capture_time = t;
end

end

% Updating the time
t = t + dt;

step = step + 1;
end

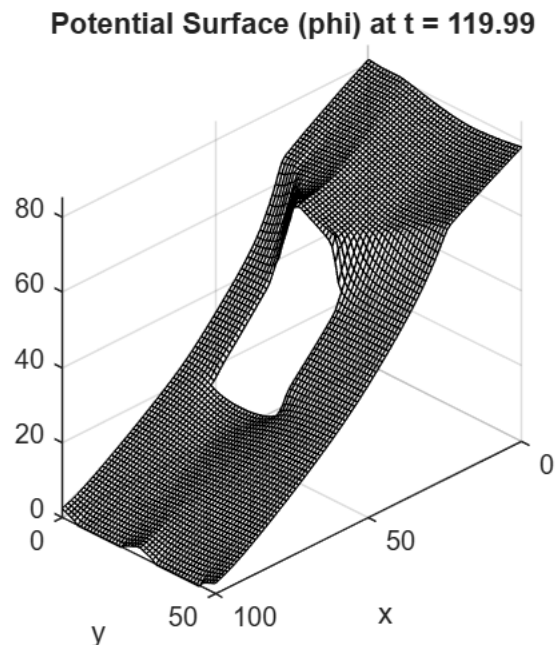
% Close both Videos
% ---- Finalize videos (REQUIRED – flushes frames to disk, finalizes the .mp4
container) ----
close(vContour);
close(vSurface);
close(vVelocity);

```

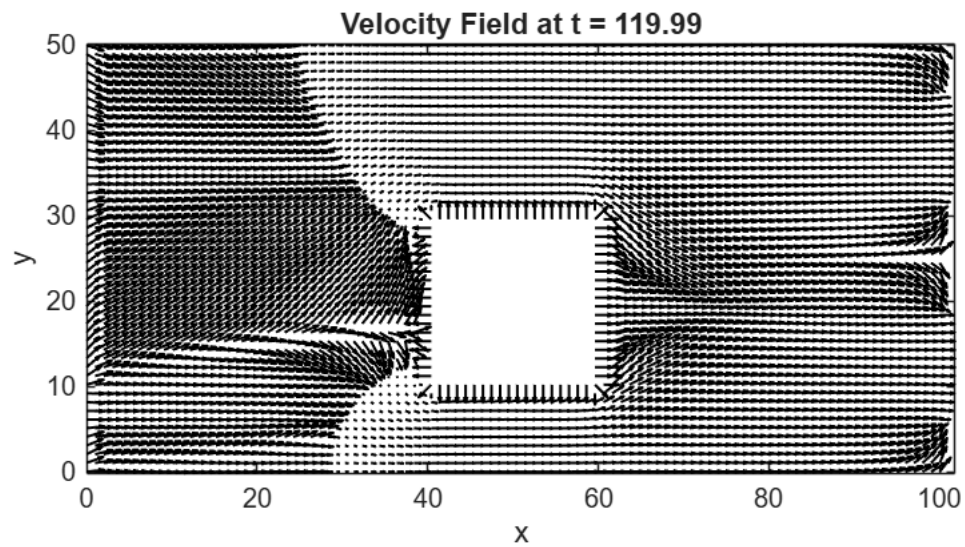
```
% ---- Close figure windows (optional cleanup) ----
close(figContour);
close(figSurface);
close(figVelocity);
fprintf("Capture time = %.2f\n", capture_time);
```

```
% Mask obstacle / un-reached cells before plotting
phi_plot = phi_at_120;
phi_plot(phi_plot > 1e6) = NaN; % anything near INFINITE gets excluded

figure(10);
surf(x_plot, y_plot, phi_plot, 'EdgeColor', 'black', 'FaceColor', 'none');
view(45+90, 30);
axis equal;
zlim([0 max(phi_plot(:), [], 'omitnan')*1.05]); % optional: pin z-range to real
data
title(sprintf('Potential Surface (phi) at t = %.2f', capture_time));
xlabel('x'); ylabel('y');
```



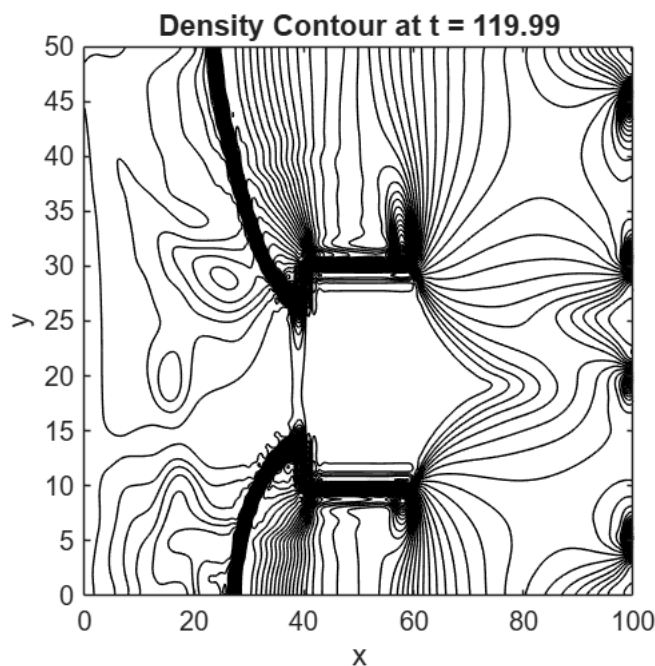
```
% Plotting the Velocity Field (quiver) at t = 120s
figure(11);
quiver(x_plot, y_plot, Vx_at_120, Vy_at_120, 1.5, 'Color', 'k', 'LineWidth', 1);
xlim([0 Lx]);
ylim([0 Ly]);
title(sprintf('Velocity Field at t = %.2f', capture_time));
xlabel('x'); ylabel('y');
axis equal tight;
```



```

%% Plotting the Density Contour (black & white, line contour) at t = 120s
figure(12);
contour(Xq, Yq, rho_fine, 60, 'LineColor', 'black'); % 60 contour levels instead
of 20
xlim([0 Lx]);
ylim([0 Ly]);
title(sprintf('Density Contour at t = %.2f', capture_time));
xlabel('x'); ylabel('y');
pbaspect([1 1 1]);

```



```

% Plot of density at x = 36 m
figure(8);
plot(y_plot, rho_at_36, '-', 'LineWidth', 1);
hold on;
%%Import CSV file
data = readtable('Density_At_36_Paper_Result.csv');

% Extract columns from CSV
y_csv = data.y;
rho_csv = data.rho;
% Plot CSV data on same graph
plot(y_csv, rho_csv, 'ro', 'MarkerSize', 5);

ylim([-1 10]);

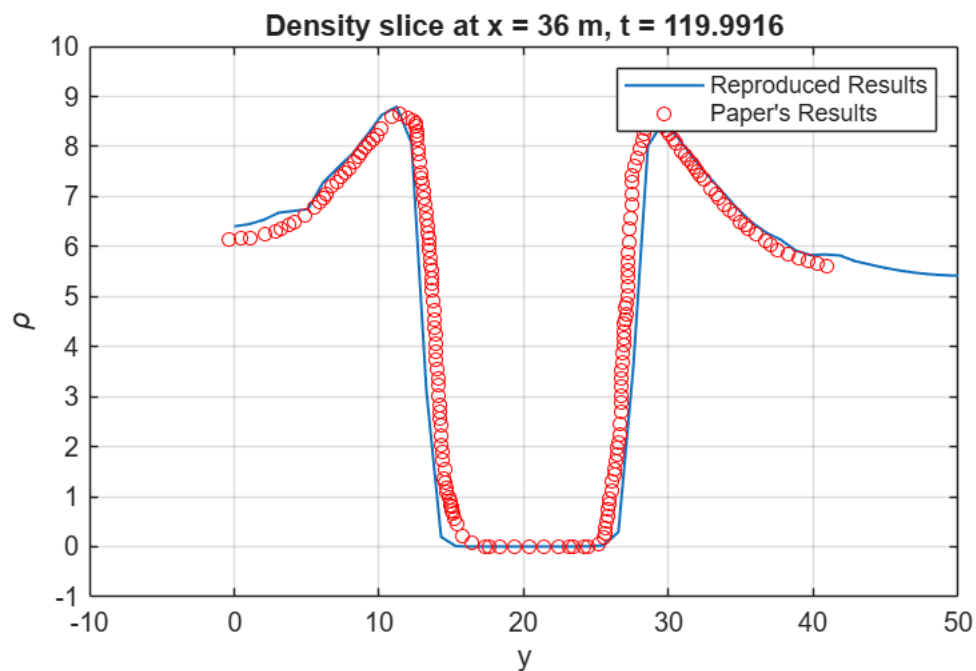
xlabel('y');
ylabel('\rho');

title(['Density slice at x = 36 m, t = ', num2str(capture_time)]);

legend("Reproduced Results", "Paper's Results");

grid on;
hold off;

```



```

%% Plotting density at y = 33 m
% Read CSV file
data = readtable('Density_At_33_Paper_Result.csv');

```

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% Extract columns from CSV
x_csv = data.x;
rho_csv = data.rho;    % or data.y depending on which contains density

% Plot paper result
figure(9);
plot(x_plot, rho_at_33, '-', 'LineWidth', 1);
hold on;

% Plot your CSV data
plot(x_csv, rho_csv, 'ro', 'MarkerSize', 5);

ylim([-1 10]);

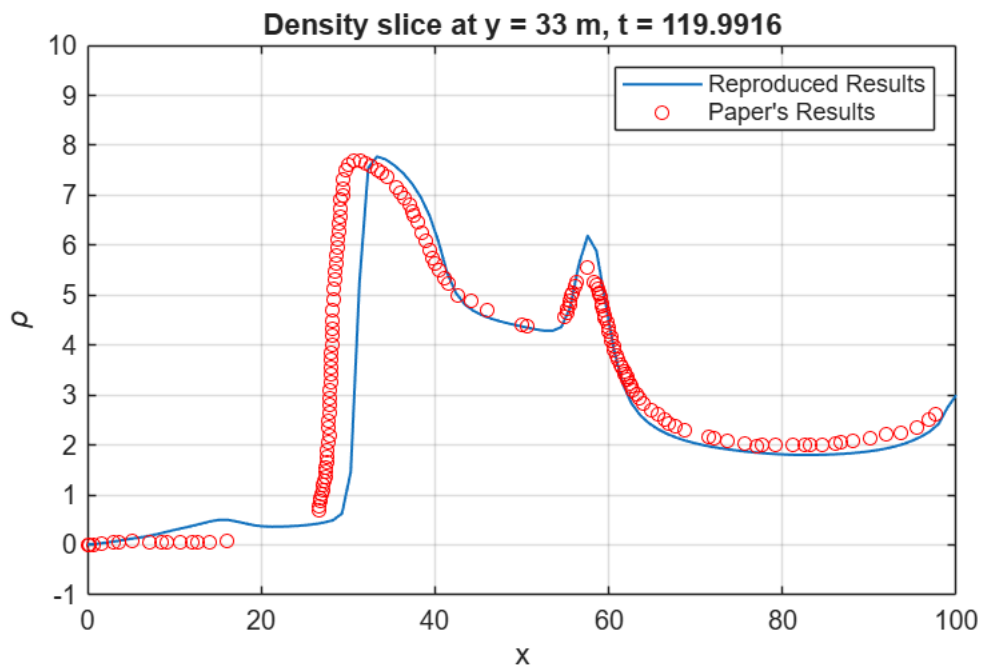
xlabel('x');
ylabel('\rho');

title(['Density slice at y = 33 m, t = ', num2str(capture_time)]);

legend("Reproduced Results", "Paper's Results");

grid on;
hold off;

```



First order Godunov Fast Sweep Algorithm (Older Version)

```

function phi = First_order_Godunov_Fast_Sweep_Algorithm(c,h,INFINITE)
    global Nx Ny obstacle;
    phi = INFINITE*ones(Ny, Nx);    % Initial potential field

    % Exit Boundary (Right boundary at the exit)

```

```

for j = 1 : Ny
    if((j >= 9 && j <= 23) || (j >= 34 && j <= 48))
        phi(j,Nx-3) = 0;
    end
end

%-----
% Fast Sweeping Iterations
%-----

for sweep = 1:20

    % Sweep 1:
    for i = 4:Nx-2
        for j = 4:Ny-2
            if ~obstacle(j, i)
                phi = update_phi(phi, c, i, j, h);
                if i == Nx-3 && ((j >= 9 && j <= 23) || (j >= 34 && j <= 48))
                    phi(j,i) = 0;
                end
            else
                phi(j,i) = INFINITE;
            end
        end
    end

    % Sweep 2:
    for i = Nx-1:-1:2
        for j = 2:Ny-1
            if ~obstacle(j, i)
                phi = update_phi(phi, c, i, j, h);
                if i == Nx-3 && ((j >= 9 && j <= 23) || (j >= 34 && j <= 48))
                    phi(j,i) = 0;
                end
            else
                phi(j,i) = INFINITE;
            end
        end
    end

    % Sweep 3:
    for i = Nx-1:-1:2
        for j = Ny-1:-1:2
            if ~obstacle(j, i)
                phi = update_phi(phi, c, i, j, h);
                if i == Nx-3 && ((j >= 9 && j <= 23) || (j >= 34 && j <= 48))
                    phi(j,i) = 0;
                end
            else
                phi(j,i) = INFINITE;
            end
        end
    end
end

```

```

        end
    end

    % Sweep 4:
    for i = 2:Nx-1
        for j = Ny-1:-1:2
            if ~obstacle(j, i)
                phi = update_phi(phi, c, i, j, h);
                if i == Nx-3 && ((j >= 9 && j <= 23) || (j >= 34 && j <= 48))
                    phi(j,i) = 0;
                end
            else
                phi(j,i) = INFINITE;
            end
        end
    end

    % Extrapolation of the phi values

    % Bottom Edge    (3 Ghost Points)
    phi(3,:) = phi(4,:);
    phi(2,:) = phi(3,:);
    phi(1,:) = phi(2,:);

    % Top Edge      (3 Ghost Points)
    phi(Ny-2,:) = phi(Ny-3,:);
    phi(Ny-1,:) = phi(Ny-2,:);
    phi(Ny,:) = phi(Ny-1,:);

    % Right Edge    (3 Ghost Points)
    for j = 1:Ny
        if (j >= 9 && j <= 23) || (j >= 34 && j <= 48)    % exit rows: linear
            extrapolation
                slope = phi(j,Nx-3) - phi(j,Nx-4);
                phi(j,Nx-2) = phi(j,Nx-3) + slope;
                slope = phi(j,Nx-4) - phi(j,Nx-5);
                phi(j,Nx-1) = phi(j,Nx-2) + slope;
                slope = phi(j,Nx-5) - phi(j,Nx-6);
                phi(j,Nx) = phi(j,Nx-1) + slope;
            else
                % wall rows: keep
                flat copy
                    phi(j,Nx-2) = phi(j,Nx-3);
                    phi(j,Nx-1) = phi(j,Nx-2);
                    phi(j,Nx) = phi(j,Nx-1);
                end
            end
        end

    % Left Edge      (3 Ghost Points)
    phi(:,3) = phi(:,4);
    phi(:,2) = phi(:,3);

```

```

        phi(:,1) = phi(:,2);
    end
end

% Subfunction: Update one grid point using Godunov scheme

function phi = update_phi(phi, c, i, j, dx)
    a = min(phi(j, i-1), phi(j, i+1)); % x-direction neighbors
    b = min(phi(j-1, i), phi(j+1, i)); % y-direction neighbors
    c_i = c(j, i); % Local inverse speed
    if abs(a - b) >= c_i * dx
        phi(j, i) = min(a, b) + c_i * dx;
    else
        inside = 2 * (c_i * dx)^2 - (a - b)^2;
        if inside >= 0
            phi(j, i) = (a + b + sqrt(inside)) / 2;
        end
    end
end
end

```

WENO FAST SWEEP

```

function phi =
weno_fast sweep(c,h,eta,error_eta,INFINITE,initial_guess,iteration_error)

    global Nx Ny obstacle;

    % Phi is initialized with infinity
    phi = initial_guess;
    phi_old = INFINITE * ones(size(phi));

    % Iterations
    iterations = 0;
    loop_safety = 0;

    % Iterations
    while (sum(abs(phi(:)-phi_old(:))) > error_eta && loop_safety<1000)

        %fprintf("Iteration = %d ",iterations);
        err = sum(abs(phi(:)-phi_old(:)));
        %fprintf("Error = %0.10f\n",err);

        phi_old = phi;
        iterations = iterations + 1;
        iteration_error(iterations) = err;
        loop_safety = loop_safety + 1;

        for sweep = 1:4
            switch sweep

```

```

case 1
    ix = 4:1:Nx-3; jy = 4:1:Ny-3;
case 2
    ix = 4:1:Nx-3; jy = Ny-3:-1:4;
case 3
    ix = Nx-3:-1:4; jy = 4:1:Ny-3;
case 4
    ix = Nx-3:-1:4; jy = Ny-3:-1:4;
end

for i = ix
    for j = jy
        %if the point is outside the obstacle
        if(~obstacle(j,i))
            %-----Calculating the phix_min-----_%

            r_back = (eta + (phi(j,i) - 2*phi(j,i-1) + phi(j,i-2))^2)/
(eta + (phi(j,i+1) - 2*phi(j,i) + phi(j,i-1))^2);
            r_front = (eta + (phi(j,i) - 2*phi(j,i+1) + phi(j,i+2))^2)/
(eta + (phi(j,i+1) - 2*phi(j,i) + phi(j,i-1))^2);

            % Calculate the w
            w_front = 1/(1+2*(r_front^2));
            w_back = 1/(1+2*(r_back^2));

            % Dell phi by Dell x plus and minus
            phix_minus = (1-w_back)*((phi(j,i+1)-phi(j,i-1))/(2*h)) +
w_back*((3*phi(j,i) - 4*phi(j,i-1) + phi(j,i-2))/(2*h));
            phix_plus = (1-w_front)*((phi(j,i+1)-phi(j,i-1))/(2*h)) +
w_front*((-3*phi(j,i)+4*phi(j,i+1)-phi(j,i+2))/(2*h));

            phix_min = min((phi(j,i) - h*phix_minus),(phi(j,i) +
h*phix_plus));

            %-----calculating the phiy_min-----_%

            r_back = (eta + (phi(j,i) - 2*phi(j-1,i) + phi(j-2,i))^2) /
(eta + (phi(j+1,i) - 2*phi(j,i) + phi(j-1,i))^2);
            r_front = (eta + (phi(j,i) - 2*phi(j+1,i) +
phi(j+2,i))^2) / (eta + (phi(j+1,i) - 2*phi(j,i) + phi(j-1,i))^2);

            % Calculate the w
            w_front = 1 / (1 + 2*(r_front^2));
            w_back = 1 / (1 + 2*(r_back^2));

            % Dell phi by Dell x plus and minus
            phiy_minus = (1-w_back) * ((phi(j+1,i)-phi(j-1,i))/(2*h)) +
w_back * ((3*phi(j,i) - 4*phi(j-1,i) + phi(j-2,i))/(2*h));

```

```

        phiy_plus = (1-w_front) * ((phi(j+1,i) - phi(j-1,i))/(2*h))
+ w_front * ((-3*phi(j,i) + 4*phi(j+1,i) - phi(j+2,i))/(2*h));

        phiy_min = min((phi(j,i) - h*phiy_minus), (phi(j,i) +
h*phiy_plus));

        if abs(phix_min - phiy_min) >= c(j,i)*h
            phi(j,i) = min(phiy_min,phix_min) + c(j,i)*h;
        else
            phi(j,i) = ((phix_min + phiy_min) + (2*(c(j,i)*h)^2 -
(phix_min - phiy_min)^2)^0.5)/2;
        end

        % Enforce the value of phi to be zero at some parts of the
right boundary
        if(i == Nx-3)
            if((j >= 9 && j <= 23) || (j >= 34 && j <= 48))
                phi(j,i) = 0;
            end
        end

        % If the point is inside the obstacle
        else
            phi(j,i) = INFINITE;
        end
    end
end
end
end
end
end
end

```

RHS of Semi Discretize Form

```

function semi_discrete_form = makeRHS(rho,t,speed)
    global Nx Ny x_velocity y_velocity h obstacle;
    semi_discrete_form = zeros(Ny-6,Nx-6);

    %--Flux for the given rho distribution
    flux = makeFlux(rho,x_velocity,y_velocity);
    flux_x = squeeze(flux(1,:,:));
    flux_y = squeeze(flux(2,:,:));

    % Calculating the maximum speed (multiplying by 1.5 as suggested)
    % alpha_x = max(abs(x_velocity(:))) * 1.5;
    % alpha_y = max(abs(y_velocity(:))) * 1.5;
    alpha_x = max(speed(:));
    alpha_y = max(speed(:));

```

```

% -- Calculating the value of f+ and f- at {i+2,i+1,i,i-1,i-2,i-3} -- %
fx_negative_whole = zeros(Ny,Nx);
fx_positive_whole = zeros(Ny,Nx);
fy_negative_whole = zeros(Ny,Nx);
fy_positive_whole = zeros(Ny,Nx);

for j = 1:1:Ny
    for i=1:1:Nx
        fx_positive_whole(j,i) = 1/2 * (flux_x(j,i) + alpha_x*rho(j,i));
        fx_negative_whole(j,i) = 1/2 * (flux_x(j,i) - alpha_x*rho(j,i));
        fy_positive_whole(j,i) = 1/2 * (flux_y(j,i) + alpha_y*rho(j,i));
        fy_negative_whole(j,i) = 1/2 * (flux_y(j,i) - alpha_y*rho(j,i));
    end
end

% -- Computing the RHS of Semi Discrete form.
for i = 4:1:Nx-3
    for j = 4:1:Ny-3
        % compute only at the points where obstacle doesn't exist
        if(~obstacle(j,i))
            %--X--%

            % fx{+}_[i+1/2,j] - Given in the paper
            fx_plus_x_half_forward = stencil( ...
                fx_positive_whole(j,i-2), ...
                fx_positive_whole(j,i-1), ...
                fx_positive_whole(j,i), ...
                fx_positive_whole(j,i+1), ...
                fx_positive_whole(j,i+2));

            % fx{+}_[i-1/2,j] - By substituting i -> (i-1) in fx{+}_[i+1/2,j]
            fx_plus_x_half_backward = stencil( ...
                fx_positive_whole(j,i-3), ...
                fx_positive_whole(j,i-2), ...
                fx_positive_whole(j,i-1), ...
                fx_positive_whole(j,i), ...
                fx_positive_whole(j,i+1));

            % fx{-}_[i+1/2,j] ( mirror of fx{+}_[i+1/2,j] )
            fx_minus_x_half_forward = stencil( ...
                fx_negative_whole(j,i+3), ...
                fx_negative_whole(j,i+2), ...
                fx_negative_whole(j,i+1), ...
                fx_negative_whole(j,i), ...
                fx_negative_whole(j,i-1));

            % fx{-}_[i-1/2,j] ( mirror of fx{+}_[i-1/2,j] )
            fx_minus_x_half_backward = stencil( ...
                fx_negative_whole(j,i+2), ...

```

```

        fx_negative_whole(j,i+1), ...
        fx_negative_whole(j,i), ...
        fx_negative_whole(j,i-1), ...
        fx_negative_whole(j,i-2));

%--Y--%

% fy{+}_[i,j+1/2]
fy_plus_y_half_forward = stencil( ...
    fy_positive_whole(j-2,i), ...
    fy_positive_whole(j-1,i), ...
    fy_positive_whole(j,i), ...
    fy_positive_whole(j+1,i), ...
    fy_positive_whole(j+2,i));

% fy{+}_[i,j-1/2]
fy_plus_y_half_backward = stencil( ...
    fy_positive_whole(j-3,i), ...
    fy_positive_whole(j-2,i), ...
    fy_positive_whole(j-1,i), ...
    fy_positive_whole(j,i), ...
    fy_positive_whole(j+1,i));

% fy{-}_[i,j+1/2]
fy_minus_y_half_forward = stencil( ...
    fy_negative_whole(j+3,i), ...
    fy_negative_whole(j+2,i), ...
    fy_negative_whole(j+1,i), ...
    fy_negative_whole(j,i), ...
    fy_negative_whole(j-1,i));

% fy{-}_[i,j-1/2]
fy_minus_y_half_backward = stencil( ...
    fy_negative_whole(j+2,i), ...
    fy_negative_whole(j+1,i), ...
    fy_negative_whole(j,i), ...
    fy_negative_whole(j-1,i), ...
    fy_negative_whole(j-2,i));

    fx_positive_half = fx_plus_x_half_forward + fx_minus_x_half_forward;
    fx_negative_half = fx_plus_x_half_backward +
fx_minus_x_half_backward;
    fy_positive_half = fy_plus_y_half_forward + fy_minus_y_half_forward;
    fy_negative_half = fy_plus_y_half_backward +
fy_minus_y_half_backward;

%--Boundary Condition -- %

% Left Boundary (Flux need not be computed)

```

```

if(i == 4)
    % Y flux is said to be zero at the inflow
    fy_negative_half = 0;
    % Unsteady Boundary Condition (for the x flux)
    if(t >= 0 && t <= 60)
        fx_negative_half = t/12;
    elseif(t >= 60 && t <= 120)
        fx_negative_half = 10 - t/12;
    elseif(t >= 120)
        fx_negative_half = 0;
    end
end

% Bottom Wall (Normal Flux on the boundary is zero)
if(j == 4)
    fy_negative_half = 0;
end

% Top Wall (Normal Flux on the boundary is zero)
if(j == Ny-3)
    fy_positive_half = 0;
end

% Right Boundary (Normal Flux at non-exit is zero)
if(i == Nx-3)
    % At the Walls
    if(~((j >= 9 && j <= 23) || (j >= 34 && j <= 48)))
        fx_positive_half = 0;
    % At the Outflow
    else
        % Extrapolation of X and Y flux will be done at the
boundary.

        fx_positive_half = flux_x(j, Nx-3);
        fy_positive_half = flux_y(j, Nx-3);
    end
end

% At the Boundary of the obstacle

if(j >= 14 && j <= 33)
    if(i == 43)
        fx_positive_half = 0; % Left Boundary
    elseif(i == 64)
        fx_negative_half = 0; % Right Boundary
    end
elseif(i <= 63 && i >= 44)
    if(j == 13)
        fy_positive_half = 0; % Bottom Boundary
    elseif(j == 34)
        fy_negative_half = 0; % Top Boundary
    end
end
end

```

```

        % Computing RHS of Semi Discretized form
        semi_discrete_form(j-3,i-3) = - 1/h * (fx_positive_half -
fx_negative_half) - 1/h * (fy_positive_half - fy_negative_half);

    else
        % Else when there is an obstacle

    end

end

end

end
end

```

WENO Advection Stencil

```

% To compute f half. Order of input matters
function f = stencil(f1,f2,f3,f4,f5)

    eta = 10^-6;

    % Compute S values
    s_1 = ((1/3) * f1) + ((-7/6) * f2) + ((11/6) * f3);
    s_2 = ((-1/6) * f2) + ((5/6) * f3) + ((1/3) * f4);
    s_3 = ((1/3) * f3) + ((5/6) * f4) + ((-1/6) * f5);

    % Compute Beta
    beta_1 = 13/12 * (f1 - 2*f2 + f3)^2 + 1/4 * (f1 - 4*f2 + 3*f3)^2;
    beta_2 = 13/12 * (f2 - 2*f3 + f4)^2 + 1/4 * (f2 - f4)^2;
    beta_3 = 13/12 * (f3 - 2*f4 + f5)^2 + 1/4 * (3*f3 - 4*f4 + f5)^2;

    % Defining Gamma
    gamma_1 = 1/10;
    gamma_2 = 3/5;
    gamma_3 = 3/10;

    % Computing weights bar
    w_1_dash = ( gamma_1 ) ./ ((eta + beta_1)^2);
    w_2_dash = ( gamma_2 ) ./ ((eta + beta_2)^2);
    w_3_dash = ( gamma_3 ) ./ ((eta + beta_3)^2);

    % Computing Weights
    w_1 = ( w_1_dash ) / (w_1_dash + w_2_dash + w_3_dash);
    w_2 = ( w_2_dash ) / (w_1_dash + w_2_dash + w_3_dash);
    w_3 = ( w_3_dash ) / (w_1_dash + w_2_dash + w_3_dash);

    % Computing Flux
    f = w_1 * s_1 + w_2 * s_2 + w_3 * s_3;

end

```

